

INSTITUTE OF COMPUTER SCIENCE

INTRODUCTION

The Department of Computer Science was established as a full-fledged department in 1999 and recently, in 2021, has been upgraded as Institute of Computer Science, with the goal to offer high quality degree programs that ensure that students will be able to integrate theory and practice recognize the importance of abstraction and appreciate the value of efficient design created to meet clearly developed requirements. The institute intends to prepare students for lifelong learning as they undertake professional careers in computing. The institute has always close ties to the other departments of Physical Sciences/Natural Sciences of the University, even has increasingly experienced collaborations with all other disciplines/faculties outside Pakistan important to the University. It is through these collaborations that the importance of Computer Science in a broader sense is best appreciated.

To be a leading Institution of Computer Science dedicated to fostering innovation, excellence, and ethical practices in computing for the betterment of society and the advancement of knowledge.

MISSION

Our mission is to provide high-quality education, research, and practical experiences in Computer Science, equipping students with the skills, knowledge, and ethical values necessary to excel in diverse computing fields. We are committed to promoting a culture of collaboration, critical thinking, and lifelong learning, while actively engaging with industry, academia, and the community to address local and global challenges through computing solutions.

PROGRAMS OBJECTIVES

- 1) Develop a comprehensive understanding of core concepts, principles, and theories in computer science, aligned with industry trends and technological advancements.
- 2) Cultivate analytical, problem-solving, and critical thinking skills necessary to design, develop, and implement innovative computing solutions to address real-world challenges.
- 3) Foster creativity and innovation in students to encourage the exploration and application of emerging technologies for the advancement of computer science and its interdisciplinary applications.
- 4) Promote ethical behavior, professionalism, and social responsibility among students, emphasizing the importance of integrity, inclusivity, and respect for diversity in the computing profession.
- 5) Provide practical experiences, project based learning internships, and industry collaborations to enhance students' readiness for successful careers in diverse fields of Computer Science.
- 6) Encourage lifelong learning and professional development through continuous education, research opportunities, and engagement with academic, industry, and community partners.
- 7) Contribute to the socioeconomic development of society by leveraging computing expertise to address local and global challenges and improve the quality of life for individuals and communities.

PROGRAMS OFFERED

- BS (CS) (Four-Year)
- BS (IT) (Four-Year)
- BS (SE) (Four-Year)
- MS/MPhil (CS)
- PhD

SCHEME OF STUDIES FOR BS (CS) (FOUR-YEAR PROGRAM)

BS (CS) Part-I Semester-I

Course No#	Course Title	Credit Hours
CS-101	Programming Fundamentals	4 (3+1)
GE-101	Introduction to Computing	3 (2+1)
GE-102	Discrete Structures	3
GE-103	Islamic Studies	2
GE-104	English Composition & Comprehension	3
GE-105	Calculus & Analytical Geometry	3
GE-106	Understanding of Holy Quran –I/ Ethic (Non Muslims)	1
	Total	18 (16+2)

Semester-II

Course No#	Course Title	Credit Hours
CS-102	Object Oriented Programming	4 (3+1)
CS-103	Digital Logic Design	3 (2+1)
GE-106	Introduction to Management	2
GE-107	Ideology and Constitution of Pakistan	2
GE-108	Expository Writing	3
MT-102	Multivariable Calculus	3
GE-109	Understanding of Holy Quran – I/ Ethic (Non Muslims)	1
	Total	18 (15+2)

BS (CS) Part-II

Semester-III

Course No#	Course Title	Credit Hours
CS-201	Data Structures	4 (3+1)
CS-202	Computer Networks	3 (2+1)
GE-201	Applied Physics	3 (2+1)
GE-202	Entrepreneurship	2
GE-203	Civics and Community Engagement	2
MT-201	Statistics & Probability	3
	Total	17 (14 + 3)

Semester-IV

Course No#	Course Title	Credit Hours
CS-203	Operating Systems	3 (2+1)
CS-204	Information Security	3 (2+1)
CS-205	Software Engineering	3 (2+1)
CS-206	Database Systems	4 (3+1)
GE-204	Professional Practices	2
MT-202	Linear Algebra	3
	Total	18 (14 + 4)

BS (CS) Part-III

Semester-V

Course No#	Course Title	Credit Hours
CS-301	Theory of Automate	3
CS-302	Analysis of Algorithms	3
CS-303	Computer Organization & Assembly Language	3 (2+1)
CS-304	Web Technologies	3 (2+1)
CS-305	Computer Architecture	3 (2+1)
EN-301	Technical & Business Writing	3
	Total	18 (15 + 3)

Semester-VI

Course No#	Course Title	Credit Hours
CS-306	Artificial Intelligence	3 (2+1)
CS-307	Compiler Construction	3 (2+1)
CS-308	Parallel & Distributed Computing	3 (2+1)
CS-310	Mobile Application Development-I	3 (2+1)
CS-311	HCI & Computer Graphics	3 (2+1)
MG-301	Introduction to Marketing	3
	Total	18 (13+5)

BS (CS) Part-IV

Semester-VII

Course No#	Course Title	Credit Hours
CS-401	Final Year Project I	3 (0+3)
CS-402	Advanced Database Management Sys.	3 (2+1)
CS-403	Advanced Programming (Visual Programming)	3 (2+1)
CS-404	Internship	3 (0+3)
	Total	12 (4+8)

Semester-VIII

Course No#	Course Title	Credit Hours
CS-405	Final Year Project II	3 (0+3)
CS-406	Digital Image Processing	4 (3+1)
CS-407	Gaming Animation	3 (2+1)
CS-408	Machine Learning	3 (2+1)
	Total	13 (7+6)

Semester-VI

Course No#	Course Title	Credit Hours
CS-307	Artificial Intelligence	3 (2+1)
CS-308	Cyber Securities	3 (2+1)
CS-309	Parallel & Distributed Computing	3 (2+1)
CS-310	Mobile Application Development-I	3 (2+1)
CS-311	Human Computer Interaction	3 (2+1)
MG-311	Introduction to Marketing	3
	Total	18 (13+5)

BS (IT) Part-IV

Semester-VII

Course No#	Course Title	Credit Hours
CS-401	Final Year Project I	3 (0+3)
CS-402	Database Admin. & Management	3 (2+1)
CS-403	System & Network Administration	3 (2+1)
CS-404	Internship	3 (0+3)
	Total	12 (4+8)

Semester-VIII

Course No#	Course Title	Credit Hours
CS-405	Final Year Project II	3 (0+3)
CS-406	Information Technology Infrastructure	3 (2+1)
CS-407	Network Security	3 (2+1)
CS-407	Software Project Management	3 (2+1)
	Total	12 (6+6)

SCHEME OF STUDIES FOR BS (SE) (FOUR-YEAR PROGRAM)

BS (SE) Part-I
Semester-I

Course No#	Course Title	Credit Hours
CS-101	Programming Fundamentals	4(3-3)
GE-101	Application of Information & Communication Technologies	3(2-3)
GE-102	Discrete Structures	3(3-0)
GE-103	Calculus and Analytic Geometry	3(3-0)
GE-104	Functional English	3(3-0)
GE-105	Understanding of Holy Quran –I/ Ethic (Non Muslims)	1
	Total	17(14-6)

Semester-II

Course No#	Course Title	Credit Hours
CS-102	Object Oriented Programming	4(3-3)
CS-103	Database Systems	4(3-3)
CS-104	Digital Logic Design	3(2-3)
MT-101	Multivariable Calculus	3(3-0)
MT-102	Linear Algebra	3(3-0)
GE-106	Understanding of Holy Quran –II/ Ethic (Non Muslims)	1
	Total	18(14-9)

BS (SE) Part-II
Semester-III

Course No#	Course Title	Credit Hours
CS-201	Data Structures	4(3-3)
CS-202	Information Security	3(2-3)
CS-203	Artificial Intelligence	3(2-3)
CS-204	Computer Networks	3(2-3)
SE-201	Software Engineering	3(3-0)
MT-201	Probability & Statistics	3(3-0)
	Total	19(15-12)

Semester-IV

Course No#	Course Title	Credit Hours
CS-205	Computer Organization & Assembly Language	3(2-3)
SE-202	Software Design & Architecture	3(3-0)
SE-203	Software Construction & Development	3(2-3)
GE-201	Applied Physics	3(2-3)
GE-202	Expository Writing	3(3-0)
GE-203	Islamic Studies	2(2-0)
	Total	17(14-9)

BS (SE) Part-III
Semester-V

Course No#	Course Title	Credit Hours
CS-301	Operating Systems	3(2-3)
SE-301	Software Quality Engineering	3(2-3)
SE-302	Software Requirement Engineering	3(2-3)
SE-303	Software Verification and Validation	3(2-3)
SE-304	Object Oriented Analysis & Design	3(2-3)
GE-301	Introduction to Management	2(2-0)
	Total	17(12-15)

Semester-VI

Course No#	Course Title	Credit Hours
SE-305	Software Project Management	3(2-3)
SE-306	Parallel & Distributed Computing	3(2-3)
CS-302	Computer Architecture	3(2-3)
CS-303	Theory of Automata	3(2-3)
CS-304	HCI & Computer Graphics	3(2-3)
CS-305	Web Technologies	3(2-3)
	Total	18(12-18)

BS (SE) Part-IV

Semester-VII

Course No#	Course Title	Credit Hours
CS-401	Final Year Project-I	2(0-6)
CS-402	Analysis of Algorithms	3(3-0)
CS-403	Advanced Database Management	3(2-3)
SS-401	Introduction to Marketing	3(3-0)
EN-401	Technical & Business Writing	3(3-0)
GE-401	Entrepreneurship	2(2-0)
	Total	16(13-9)

Semester-VIII

Course No#	Course Title	Credit Hours
CS-404	Final Year Project-II	4(0-12)
GE-402	Ideology and Constitution of Pakistan	2(2-0)
GE-403	Professional Practices	2(2-0)
GE-404	Civics and Community Engagement	2(2-0)
	Total	10(6-12)